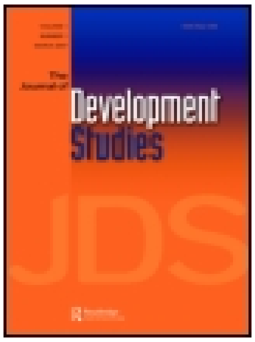

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# Informal Employment in Developing Economies: Multiple Heterogeneity

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# Informal Employment in Developing Economies: Multiple Heterogeneity

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ABSTRACT *This paper contributes to the literature on the nature of informal employment in developing economies. Drawing on the model with essential heterogeneity, it offers a list of scenarios describing the behavioural patterns which informal workers follow. The list nests not only classical patterns of a rationed formal sector versus an integrated labour market, but also different patterns of rationing. Using non-parametric techniques and data from a few African economies with different levels of development, the paper proposes empirical case studies fitting various informality schemes. Developing economies show disparate patterns of allocation of workers and various patterns of rationing.*



## 1. Introduction

A fundamental question related to labour regulations in the context of informal labour is the nature of informal employment and the mechanisms driving expansion or reduction of informality. The question is particularly important in developing economies, where the phenomenon of informality is widespread and its mechanisms have engendered debate in the literature for several decades. The most recent literature suggests heterogeneity of informal employment in developing economies (Bargain & Kwenda, 2014; Cunningham & Maloney, 2001; Dimova, Nordman, & Roubaud, 2010; Fields, 1990; Günther & Launov, 2012; Radchenko, 2014). Heterogeneity implies here that informal employment consists of both a tier that is constrained to hold an informal rather than formal job and a tier that chooses informal employment. The first tier fits the dual market model assuming two distinct segments of the labour market: a formal segment paying high wages and providing job security as well as social security and an informal sector paying low wages and serving as a last resort to unemployment (Fields, 2005; Harris & Todaro, 1970; Lewis, 1954; Todaro & Smith, 2003). The second tier is associated with an individual's cost-benefit analysis and stems from the voluntary choices of workers (Maloney, 1999, 2004; Rosenzweig, 1988).

The majority of studies empirically testing the relevance of segmentation look for grouping effects indicating the presence of the two non-integrated sectors using fully parametric methods (Basu, 1997; Cunningham & Maloney, 2001; Dickens & Lang, 1985; Dimova et al., 2010; Günther & Launov, 2012; Heckman & Hotz, 1986; Magnac, 1991; Pratap & Quintin, 2006). Among the main findings are heterogeneity of workers in terms of human capital, unobserved skills and different returns to human capital in different segments of the labour market. Using Egyptian household survey data, Radchenko

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(2014) suggests an innovative method of analysis by drawing on the model with essential heterogeneity and the non-parametric estimation techniques proposed by Heckman, Urzua, and Vytlacil (2006). In addition to identifying differences in returns to the observable human capital and unobservable skills of formal and informal workers, the model also allows for individual differences in returns to unobservable characteristics from formal versus informal employment. Radchenko (2014) shows that the distribution of individual differences in unobservable returns from holding a formal rather than an informal job and its relationship to the allocation process between formal and informal employment can be associated with various mechanisms driving informal employment.

This paper builds on Radchenko (2014) by extending it in several ways. First, it frames the selection mechanisms nested in the model with essential heterogeneity in the context of informal employment and proposes a broader range of scenarios describing different behavioural patterns which informal workers follow and that are compatible with different kinds of empirical schemes allowed by the model with essential heterogeneity. Second, unlike Radchenko (2014), which focuses on lower-middle-income Egypt, the paper covers an upper-middle-income economy (South Africa) and a low-income economy (Uganda). Additionally, it discusses identifying assumptions and the distribution of exclusion variables required to identify the treatment parameters and offers a sensitivity analysis including Egypt. The different economies show varying patterns of allocation of workers between formal and informal employment. Moreover, they show various patterns of rationing.

Thus, the paper contributes to the literature by showing that in terms of workers' social welfare, efficiency and inefficiency of allocation between formal and informal employment is not necessarily associated with voluntary and involuntary engagement in the informal segment. Even in the framework of rationed formal employment, allocation of workers between the two segments can be efficient. This is different from the majority of work on the heterogeneity of informal employment (Cunningham & Maloney, 2001; Dimova et al., 2010; Fields, 1990; Günther & Launov, 2012; Radchenko, 2014) that focuses on within-economy heterogeneity and the inefficiency of informal employment without considering a variety of schemes of rationing.

The paper is organised as follows. Section 2 presents the data and discusses the main features of the African economies used in the case studies. Section 3 outlines the model with essential heterogeneity in the framework of formal and informal employment and its relationship to various treatment effects. Section 4 discusses the implications of the model in terms of efficiency of the allocation of workers into different employment groups. Section 5 discusses the main empirical results. Section 6 concludes.

## 2. Data and Descriptive Statistics

The data come from the *South Africa 2011 Quarterly Labor Force Survey* (QLFS 2011) and two waves from the Uganda National Panel Survey (UNPS 2009/2010 and UNPS 2010/2011). The QLFS is designed to be representative at the provincial level and within provinces. The sample size is 3080 primary sampling units (PSUs). It contains 42,777 households or about 100,000 individuals. The quarterly data are averaged over the year to smooth and reduce noise from measurement errors or outliers. The UNPS is nationally representative. The survey rounds over 2009–2011 cover approximately 3000 Ugandan households. The samples used are composed of individuals between 15 and 64 in age and engaged in the labour market.

The South Africa data provide an indicator of informal employment. Following the job based concept recommended by the International Labour Organization (ILO), South Africa Statistics adopts the broadest definition based on job-based concept and covering all workers in the informal sector as well as informal workers of the formal sector. The informal sector includes employees working in establishments that do not deduct income tax from their salary/wage and employers or own account workers who are not registered for either income tax or value-added tax. Informal workers of the formal sector are employees in the formal sector and persons employed in private households who are not entitled to basic benefits such as pensions or medical aid and who do not have a written contract of employment. This study uses the same definition keeping non-agricultural employment and including workers providing services in private households.[1] This is also the approach adopted in the literature on South African informal employment (Devey, Skinner, & Valodia, 2005; Valodia, Lebani, Skinner, & Devey, 2006).

The Ugandan data do not provide an indicator of informal employment but do contain information on taxation. Based on this information, informal employment is identified here as consisting of workers whose employers do not deduct or pay income tax from the salary and employers and independent workers not registered for income tax. The Ugandan data suffer from high rates of non-response: out of 5000 workers reporting their activity, about 2000 responded to the question on tax payment. Among these, less than 20 per cent have reported it twice, when questioned in both the 2009/10 and 2010/11 surveys. Thus, the resulting sample in use does not have a panel dimension but rather represents pooled cross-sections.

### *2.1. Labour Market Features*

Tables 1 and 2 show the distribution of the working-age population between employed, unemployed and inactive groups in the original QLFS and UNPS samples respectively. The tables also display the sample distribution of demographic characteristics and education across different groups and mean wages in the formal and informal sectors.

The unemployed and inactive South African working-age population are younger than the employed population. The average education of the non-working population is the same as that of informal employment (10–12 years) but formal employment consists of more highly educated workers. In contrast with informal and unemployed workers, white workers make up a significant fraction of the formal group. Online Appendix A describes worker's distribution by industries.

**Table 1.** Breakdown of the labour force and individual characteristics, South Africa, QLFS 2011

| | Formal | Informal | Unemployed | Inactive |
|---|---|---|---|---|
| Breakdown of the working-age population, % | 39 | | 13 | 48 |
| Non-agricultural employment | 24905 (65%) | 13410 (35%) | | |
| Discouraged/not available for work, % | | | | 70 /30 |
| Age, mean years | 38 | 38 | 30 | 31 |
| Female, % | 46 | 54 | 52 | 61 |
| Urban, % | 81 | 63 | 72 | 50 |
| Married, % | 54 | 42 | 25 | 26 |
| African, % | 64 | 87 | 85 | 85 |
| Coloured and Asian, % | 20 | 10 | 13 | 10 |
| White, % | 16 | 3 | 2 | 5 |
| Number of years of education, mean | 14 | 10 | 12 | 10 |
| Ln of hourly wage, mean | 3.191 | 2.293 | | |

**Table 2.** Breakdown of the labour force and individual characteristics, Uganda, UNPS 2009–2011

| | Formal | Informal | Unemployed | Inactive |
|---|---|---|---|---|
| Distribution of working-age population | 70 | | 6 | 24 |
| Employment breakdown | 497 (17%) | 2487 (83%) | | |
| Age, mean years | 39 | 34 | 26 | 29 |
| Female, % | 37 | 41 | 57 | 57 |
| Urban, % | 58 | 38 | 65 | 21 |
| Married, % | 77 | 55 | 30 | 43 |
| Number of years of education | 11 | 8 | 11 | 7 |
| Ln of hourly wage, mean | 9.5 | 8.43 | | |

Uganda is an interesting example of a low-income country. It represents a weakly urbanised economy with an unemployment rate of 4.2 per cent (World Bank, 2010). The bulk of labour is involved in low value-added activities, particularly in the informal and household enterprise sectors. In contrast with the South African labour market which shows a rather equal gender distribution among the two employment groups, Ugandan employment in both groups is primary male. Age and education distributions by sector follow similar patterns in both labour markets: the age distribution is relatively equal between formal and informal groups; the majority of highly-educated workers are employed formally while the majority of workers with no or low levels of education are employed informally; the higher the level of education, the greater the portion of workers in formal employment and the smaller their part in informal employment.

Figures 1 and 2 show wage distributions (in logarithm forms) in the formal and informal employment groups. Rightward shifts of the formal employment curves imply that in both labour markets, the wage distribution among formal workers is statistically superior (higher wages) to that of informal workers; a Kolmogorov–Smirnov test does not reject the hypothesis of superiority of the formal wage distributions (the *p*-values are 0.992 and 0.998 respectively).

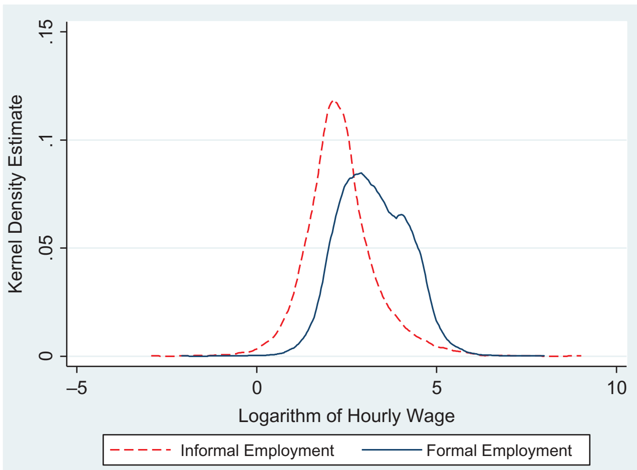


**Figure 1.** Wage distribution in South Africa.

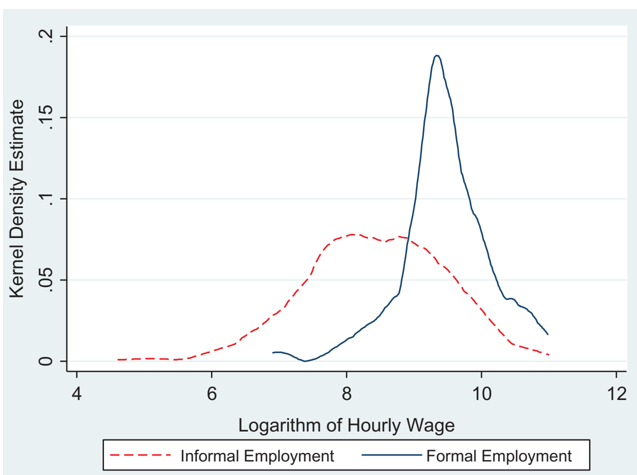


**Figure 2.** Wage distribution in Uganda.

## 3. Essential Heterogeneity: Model Outline in the Context of Formal and Informal Labour Markets

Radchenko (2014) shows how the model with essential heterogeneity (Heckman et al., 2006) applies to the framework of workers' allocation between formal and informal segments of the labour market. This section traces out the main setup detailed in Radchenko (2014) while extending it by framing the selection mechanisms that can be nested in the model. The estimation procedure and identifying assumptions are detailed in Online Appendixes B and C.

The population of workers holding formal (informal) jobs is designated as the treated (untreated) population. Depending on the worker's employment segment, either the outcome from being treated, formal earning, or the outcome from being untreated, informal earning, is observed for each worker. One's difference between formal and informal earnings is unobserved and defines one's treatment effect.

### *3.1. Base Framework*

The model with essential heterogeneity is represented by two related parts, the selection equation and the wage equation. As in Radchenko (2014), working individuals $i$ are employed either in the formal ($D_i = 1$) or informal ($D_i = 0$) employment groups based on the following selection process:

$$D_i = \begin{cases} 1, \; if \;\; F(\gamma Z_i) > U_{D,i} \\ 0, \; if \;\; F(\gamma Z_i) \leq U_{D,i} \end{cases} \tag{1}$$

where $F(\gamma Z_i)$ is the propensity score representing the probability of working formally given worker's observed characteristics $Z_i$. $F$ denotes the normal cumulative density function. $U_{D,i}$ is the unobservable counterpart of $F(\gamma Z_i)$, the propensity score of working informally based on workers' unobserved characteristics. By construction, $U_{D,i}$ is uniformly distributed over [0;1]. Now let $W_{1,i}$ and $W_{0,i}$ denote two potential outcomes corresponding to formal and informal wages. The observed and unobserved heterogeneity is modelled by allowing $W_{1,i}$ and $W_{0,i}$ to vary with observed individual wage determinants $X_i$ and unobserved individual wage determinants $U_{0,i}$,$U_{1,i}$ differing in the two segments:

$$\begin{cases} \ln W_{1,i} = \alpha_1 + \beta_1 X_i + U_{1,i}, \; if \;\; D_i = 1 \\ \ln W_{0,i} = \alpha_0 + \beta_0 X_i + U_{0,i}, \; if \;\; D_i = 0 \end{cases} \tag{2}$$

where $\alpha_0, \alpha_1, \beta_0, \beta_1$ are unknown parameters of the wage equations. $E(U_{0,i}|X_i) = 0$, $E(U_{1i}|X_i) = 0$.

The empirical model referred to below is based on the following representation of system (2):

$$\ln W_i = \alpha_0 + \beta_0 X_i + \Delta_i \cdot D_i + U_{0,i} \tag{3}$$

$\Delta_i$ here is the overall individual 'treatment' effect:

$$\Delta_i = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)X_i + (U_{1,i} - U_{0,i}) \tag{4}$$

The total treatment effect includes wage differentials related to observable individual characteristics, $(\beta_1 - \beta_0)X_i$, and unobserved treatment heterogeneity, $(U_{1,i} - U_{0,i})$ which describes essential heterogeneity, the heterogeneous difference of return from unobservable wage determinants earned if holding formal rather than informal job. Moreover, essential heterogeneity implies that $U_{D,i}$ is correlated with $U_{1,i}$, $U_{0,i}$ The correlation represents an additional flexibility of the model as compared to the class of models based on wage differentials by allowing for sorting on the gain. It is framed below.

### 3.2. Selection Mechanisms

The model outlined nests a rich set of mechanisms corresponding to different operational patterns of the labour markets hosting informal employment. This subsection explores the $U_D$ term as a function of various unobservables and provides formalisation of different context-specific selection mechanisms allowed by the model and discussed in the literature. Selection process (1) depends on workers' preferences and barriers to entry into two different employment groups. The selection determinants that are not captured by the observables are represented by the error term of the selection equation, $U_D$; they are related to the barriers to entry and the relative gains:

$$U_{D,i} = U_{D,i}\left(b_{1i}, b_0, U_{1,i} - U_{0,i}\right) \tag{5}$$

Here $b_{1i}$ stands for the unobservable determinants of entry barriers into formal employment: risk aversion to possible unemployment resulting from unsuccessful search for formal employment; entry cost such as the moving cost from rural to urban areas offering more formal jobs in the framework of the developing economy (Fields, 1990); local market and conjuncture; lack of entrepreneurial skills and background; lack of social network and environment that would favour entry. Fields (1990, 2005) points out that informal employment is not necessarily free-entry either: informal activities may also have limited access as they may require some human, physical and/or financial capital or minimal entrepreneurial skills; even unskilled activities may require access into the related network and obtaining a kind of informal license by competing/paying for running the activity. $b_0$ stands therefore for the entry barriers into informal employment.

Note that unlike $b_{1i}$, $b_{0i}$ is not individual-specific but rather $i$'s local labour force specific: indeed, the workers under consideration are successfully employed in one of the groups. Assuming that $b_{1i}$ is stronger than $b_0$ (Fields, 1990, 2005), $i$'s employment implies that at least informal employment barriers are overcome by worker $i$: $b_{0i} = 0$ for workers under consideration. $b_0$ represents, therefore, barriers faced by $i$'s local labour force. Note also that given $i$'s employment, his or her reservation wage, $w$, is not higher than the maximum of the formal and informal wages: $w \leq \max\left(\alpha_1 + \beta_1 X_i + U_{1,i},\ \alpha_0 + \beta_0 X_i + U_{0,i}\right)$. Given relationship (5), the selection mechanism can be differentiated based on different levels of $b_{1i}$, $b_0$ and their interaction with $U_{1,i} - U_{0,i}$:

(1) There are no barriers to entry in either employment group:

$$\begin{aligned} & b_{1i} = 0 \cup b_{0i} = 0 \\ & U_{D,i} = U_{D,i}\left(0, 0, U_{1,i} - U_{0,i}\right) \end{aligned} \tag{6}$$

This scheme implies that workers' allocation is guided by their relative gains or losses from working formally rather than informally. Workers' cost-benefit considerations underlie therefore the selection process in this case. The labour market is integrated with no constraints to entry from its demand side; the supply side leads the allocation to a large extent. Workers remaining unemployed in such a market are those who have a high reservation wage: $w > \max(\alpha_1 + \beta_1 X + U_1,\ \alpha_0 + \beta_0 X + U_0)$.

(2) There are no barriers to entry to informal employment but there are barriers to entry to the formal one. However, their level is correlated with the gains or losses from formal rather than informal work:

$$\begin{aligned} & b_{1i} \geq 0 \cup b_{0i} = 0 \\ & U_{D,i} = U_{D,i}\left(b_{1i}, 0, U_{1,i} - U_{0,i}\right) \\ & corr\left(b_{1i}, U_{1,i} - U_{0,i}\right) < 0 \end{aligned} \tag{7}$$

In this framework, workers may face disparate levels of barriers to entry into formal employment. More advantaged workers may have both easier entry and higher returns from formal employment. For example, higher entrepreneurial skills might favour both selection into formal employment restrained from the demand side and higher relative gains from formal work; or, more intense competing of worker *i* for formal job may be related to *i*'s higher expected wage leading to the positive sorting on the gain. The allocation process is guided by both supply and demand sides of the market. Those remaining unemployed are the workers with high reservation wage that are superior to the informal one $w > \alpha_0 + \beta_0 X + U_0$

(3) As previously, there are no barriers to entry to informal employment and there are barriers to entry to the formal one. In difference with the previous case, their levels are not correlated with the gains or losses from formal rather than informal work:

$$\begin{aligned} & b_{1i} \geq 0 \cup b_{0i} = 0 \\ & U_{D,i} = U_{D,i}\ \left(b_{1i}, 0, U_{1,i} - U_{0,i}\right) \\ & b_{1i} \perp \left(U_{1,i} - U_{0,i}\right) \end{aligned} \tag{8}$$

In this case, different employment groups offer different levels of wages for the same workers, presumably $U_{1,i} > U_{0,i}$ for an average worker. However, the allocation process is independent of the relative gains. This scheme corresponds to what is defined as dual market or segmentation (Fields, 2005). Under segmentation, employment consists of two non-competing groups, informal employment being free-entry, providing low earnings and representing a last resort; workers do not sort themselves based on anticipated individual gains. Those remaining unemployed are the workers with high reservation wage superior to the informal one: $w > \alpha_0 + \beta_0 X + U_0$.

(4) Finally, both employment groups may be restricted and have limited access. Disadvantaged workers may have a particularly low reservation wage in this context. No job insurance and lack of income assistance allowances in a developing economy context may push workers to take any job earning them some cash (Fields, 2005). The higher the barriers they face, the lower the earnings they may accept. This may induce the correlation between the barriers to entry into formal employment and relative losses from working informally rather than formally:

$$\begin{aligned} & b_{1i} \geq 0 \cup b_{0i} \geq 0 \\ & U_{D,i} = U_{D,i}\left(b_{1i}, b_{0i}, U_{1,i} - U_{0,i}\right) \\ & corr\left(b_{1i,0}, U_{1,i} - U_{0,i}\right) > 0 \end{aligned} \tag{9}$$

The unemployed population hosts in this case not only the workers with a reservation wage superior to the informal wage ($w > \alpha_0 + \beta_0 X + U_0$), but also workers unable to overcome the entry barriers even into informal employment: $w \leq \alpha_0 + \beta_0 X + U_0$ and $b_0 > 0$. The reservation wage in this framework is a function of $b_0$ (but not of $b_1$).

### *3.3. Model Outcomes*

The outcomes of the model represented by the selection Equation (1) along with wage Equations (3–4) allow us to differentiate between various patterns of formal-informal allocation discussed above and to detect a pattern underlying an observed formal-informal distribution of workers. The outcomes are detailed in Radchenko (2014) and summarised in this subsection.

The most general parameters are the average treatment effect (*ATE*) describing average gap in earnings from holding formal versus informal jobs in all employment, $E(\ln W_{1i} - \ln W_{0i})$, and average effects of treatment on the treated (*TT*) and on the non-treated (*TNT*) describing similar gaps in formal and informal employment segments respectively: $E(\ln W_{1i} - \ln W_{0i}|D_i = 1)$ and $E(\ln W_{1i} - \ln W_{0i}|D_i = 0)$. Due to additivity of model (2), the treatment effects are also additive in terms of observable and unobservable wage determinants: $ATE = ATE_X + ATE_U$, $TT = TT_X + TT_U$ and $TNT = TNT_X + TNT_U$ with:

$$\begin{aligned} ATE_X &= (\beta_1 - \beta_0)E(X_i) \\ TT_X &= (\beta_1 - \beta_0)E(X_i \mid D_i = 1) \\ TNT_X &= (\beta_1 - \beta_0)E(X_i \mid D_i = 0) \end{aligned}$$

and

$$\begin{aligned} ATE_U &= (\alpha_1 - \alpha_0) \\ TT_U &= (\alpha_1 - \alpha_0) + E(U_{1i} - U_{0i} \mid D_i = 1) \\ TNT_U &= (\alpha_1 - \alpha_0) + E(U_{1i} - U_{0i} \mid D_i = 0) \end{aligned}$$

The differences between *TT* (*TNT*) and *ATE* quantify sorting on the gain from working formally (informally) in the population. It can be also considered in terms of observable and unobservable wage determinants (Radchenko, 2014).

The *ATE, TT* and *TNT* are identified by integrating the marginal treatment effect, *MTE*, over the distribution of the probability of selecting into one group rather than another (working formally rather than informally in our context). In practice, this implies calculation of the *MTE* weighted averages (see Radchenko [2014] for the detailed presentation). The marginal treatment effect is a local parameter defining the mean marginal gross return to formal versus informal work at a specific point of $U_{D,i}$:

$$MTE(x_i, u_{D,i}) = \left.\frac{\partial E(\ln W_i | X_i = x_i, P(Z_i) = p)}{\partial p}\right|_{p=u_{D,i}} = E(\ln W_{1,i} - \ln W_{0,i} | X_i = x_i, U_{D,i} = u_{D,i})$$

$U_{D,i}$ is unobserved. However, it can be calculated at the point where worker $i$ is indifferent about working in segment 1 or segment 0. Selection process (1) implies that at this point $U_{D,i} = F(\gamma Z_i)$ and therefore $U_{D,i}$ equals the propensity score $P(Z)$ and represents the propensity to work informally. Additivity of (2) implies that *MTE* is additive in its components related to observables ($X_i$) and unobservables ($U_{D,i}$): $MTE(x_i, u_{D,i}) = (\beta_1 - \beta_0) \cdot x_i + MTE_{U,i}$ where $MTE_{U,i}$ is the marginal gap between worker's $i$ formal and informal earnings resulting from infinitesimal changes of $U_{D,i}$.

## 4. Model Implications and Efficiency of Allocation

The relationship between $MTE_U$ and $U_D$ is the leading point of the analysis. Since $U_D$ represents the propensity to work informally due to workers' unobservable factors, $MTE_U$ corresponding to its high (low) values show minimum gain or loss due to unobservable characteristics motivating informal (formal) employment rather than formal (informal) employment. More specifically, the positive (negative) $MTE_U$ is a worker's marginal gain (willingness to pay) if working formally ($D = 1$) or marginal willingness to pay (marginal gain) if working informally ($D = 0$). Radchenko (2014) suggests several model implications related to the behaviour of $MTE_U$ in relationship to $U_D$. The discussion below offers extended views on possible scenarios linking the relationships between $MTE_U$ and $U_D$ to mechanisms (6)–(9).

### 4.1. Positive Sorting on the Gain

The first scheme corresponding to the decreasing shape of $MTE_U$ in relationship to $U_D$ implies positive sorting on the gain. In this case, the average treatment effects corresponding to unobservable gaps are ordered as $TT_U > ATE_U > TNT_U$: the average gain of the formal workers from holding formal rather than informal jobs, $TT_U$, is higher than the average gain across the whole population, $ATE_U$; the average gain of the informal workers from holding formal rather than informal jobs, $TNT_U$, is the lowest in the population; specifically, a negative value of $TNT_U$ implies positive gain of the informal workers from their actual allocation into informal employment (it follows from equivalence of $E(U_1 - U_0 \mid D = 0) < 0$ and $E(U_0 - U_1 \mid D = 0) > 0$).

Positive sorting on the gain is efficient from the perspective of social welfare defined as the mean gross surplus from workers' observed employment relative to their counterfactual employment, $S_A = E\left(U_{Actual} - U_{Counterfactual}\right)$. The surplus can be presented as the integral sum of the gains from actual formal/informal status relative to the counterfactual state weighted by the probability of the current employment type:

$$S_A = \int \left(U_{Actual} - U_{Counterfactual}\right) p_{Actual} \; d\left(U_{Actual} - U_{Counterfactual}\right) \tag{10}$$

The surplus is maximised when higher positive gains $(U_{Actual} - U_{Counterfactual} > 0)$ are associated with higher probability of the observed status $p_{Actual}$ and the losses $(U_{Actual} - U_{Counterfactual} < 0)$ are zero. Radchenko (2014) associates the scheme of positive sorting on the gain with an integrated market where access is open to both formal and informal segments: workers anticipate their gains from working in one segment rather than in the other and choose the segment that provides them with higher earnings. Informal employment in such a labour market is voluntary and chosen based on comparative advantage considerations.

However, this scheme requires further investigation since it may be the net result of several alternative patterns. Indeed, while earnings distributions in the formal and informal groups have substantial overlap, $TT_U$ representing the average gain from working formally in the population of the formal workers is likely to be positive and is certainly positive under positive sorting on the gain. The sign of the $TNT_U$ representing the average gain from working formally in the population of informal workers is ambiguous and under a positive sorting on the gain may depend on prevalence of disparate tiers of informal employment and extent of freedom to entry into formal employment.

More specifically, $TT_U > ATE_U > 0 > TNT_U$ implies prevalence of upper-tier informal employment. According to Fields (1990), who introduces the term, the upper-tier consists of informal workers endowed with some financial and/or human capital. Eventually, the capital could be acquired from having held formal jobs. The related endowment may allow for developing small unregulated businesses. The upper-tier is most highly associated with voluntary engagement in informal employment. Negative $TNT_U$ implies positive average gain from working informally in the population of informal employment and therefore fits the scenario in which the labour market is to a large extent integrated as framed by selection mechanism (6), Section 3.2. This type of allocation is socially efficient: the surplus $S_A$ is positive and may be high up to its maximum.

In terms of their position relative to formal employment, the lower- tier, or 'easy entry tier' as advanced by Fields (1990), is opposed to the upper tier by its involuntary survivalist nature: the lower tier consists of the poorest workers preferring any low-paying activity to open unemployment. The presence of the lower tier having negative gain from working informally as compared to working formally might balance positive gains of the upper tier. The average $TNT_U$ in the informal population depends on the relative importance of the upper versus lower tier and the average gains and losses in the two subpopulations. Therefore, $TT_U > ATE_U > TNT_U > 0$ signals the presence of both lower and upper tiers. Weakly positive $TNT_U$ might signify a balance between them.

This scheme of rationed formal employment and a weakly developed upper tier corresponds to the second allocation pattern (7) presented in Section 3.2. In this framework, workers choose to compete or not for formal jobs based on their expected gains. The pattern is efficient from the point of view of workers' social welfare: while $S_A$ cannot attain its maximum given relative losses of lower tier of informal employment ($U_{Actual} - U_{Counterfactual} < 0$), it is strongly positive with higher gains of formal and upper tier informal workers associated with their higher probability to access employment earning them more. This pattern could be considered as interposed between an integrated market and inefficient rationing corresponding to the flat $MTE_U$ which is discussed next.

### *4.2. Zero Sorting on the Gain*

Zero correlation between $MTE_U$ and $U_D$ implies no relationship between anticipated gains from occupational choice and workers' preferences. The likely scenario in the framework of a developing labour market is constant positive $MTE_U$ yielding the same average formal-informal wage gap (due to unobservables) across the whole population of workers ($TT_U = ATE_U = TNT_U$). Such a scheme is associated with a segmented labour market: workers do not sort themselves based on anticipated individual gains as framed by (8) in Section 3.2. Segmentation here implies inefficient rationing: surplus $S_A$ is zero in the case of equal distribution of workers between formal and informal employment groups and it is negative in the case of the higher share of informal employment. We refer to this as weakly inefficient rationing to differentiate it from the strongly inefficient rationing discussed below.

### *4.3. Negative Sorting on the Gain*

The third scheme in terms of different shapes of the relationship between $MTE_U$ and $U_D$ is increasing association, implying a negative correlation between relative gains based on unobservables and the selection equation. It means that the higher the relative gains from working formally due to unobservables, the weaker the likelihood of formal work based on the same unobservables. The treatments effects are ordered as $TT_U < ATE_U < TNT_U$. Radchenko (2014) cautiously associates this pattern with the non-pecuniary comparative advantage consideration, pointing out that this pattern would fit the model only under the assumption of a negative correlation between monetary and non-monetary relative gains based on the same unobserved factors. Such a strong assumption entails weak plausibility of the prevalence of the non-pecuniary comparative advantages in a mechanism driving workers' allocations between formal and informal employment groups.

A setting which rather fits the negative sorting on the gain in the framework of a developing economy is a labour market with high unemployment and strong prevalence of the lower tier of informal employment over its upper tier. In such a setting, the lower tier informal segment may host the poorest part of the job seeker population, which uses informal employment as a last resort and accepts any wage level. A negative sorting on the gain implies here that those who are the most likely to work informally are those whose reservation wage is particularly low, making the gap between their formal counterfactual earnings and actual informal earnings particularly high. A negative sorting on the gain therefore detects allocation pattern (9) framed in Section 3.2. In terms of the social welfare of workers, this pattern suggests inefficient rationing of formal employment (surplus $S_A$ is negative since the higher losses from actual allocation are associated with their higher likelihood) and implies that workers use informal jobs as a subsistence strategy.

## 5. Results

The presentation of the results follows estimation steps summed up in Online Appendix B: selection equation estimates obtained by applying the probit model first and wage equation and differentials estimates yielded by the double residual semi-parametric regression next.

### *5.1. Selection into Formal or Informal Employment*

Both African labour markets under consideration show a progressive positive impact of education on the selection process into formal employment (first columns of Tables 3 and 4). Formal employment consists of more highly-educated workers, in line with empirical findings for developing and, in particular, African economies: for example, Radchenko (2014) for Egypt and Dimova et al. (2010) for Côte d'Ivoire, Mali, Benin, Senegal, Togo, Niger, and Burkina Faso. Gender differences in the intensity of engagement into formal employment vary across countries. In South Africa, women are more likely to work informally (6.5% marginal effect): they are mainly engaged in the agricultural and household-related work, which is more frequently informal and less remunerative (Banerjee, Galiani, Levinsohn, McLaren, & Woolard, 2008; Casale & Posel, 2002; Devey et al., 2005; Rodrik, 2008; Valodia, 2001). In Uganda they are more likely to work formally.

**Table 3.** South Africa estimates

| | Selection equation | Wage equation Informal employment | Wage equation Gap in formal-informal return |
|---|---|---|---|
| Worker's Characteristic | Coefficient (SE) | Coefficient, $\hat{\beta}_0$ (SE) | Coefficient, $(\hat{\beta}_1 - \hat{\beta}_0)$ (SE) |
| Education | | | |
| Tertiary | 1.728*** | 2.275 | −1.295** |
| | (0.066) | (0.394) | (0.464) |
| Preparatory[3] | 1.430*** | 0.423* | 0.357 |
| | (0.049) | (0.240) | (0.319) |
| Secondary | 0.649*** | −0.136 | 0.460*** |
| | (0.041) | (0.089) | (0.176) |
| Primary | 0.096** | 0.024 | 0.080 |
| | (0.043) | (0.058) | (0.147) |
| Primary Incomplete | - | - | - |
| Female | −0.222*** | −0.328*** | 0.272*** |
| | (0.016) | (0.035) | (0.049) |
| Age | −0.008*** | 0.002 | 0.006*** |
| | (0.001) | (0.002) | (0.002) |
| Job Tenure | 0.035*** | 0.013*** | −0.017*** |
| | (0.001) | (0.004) | (0.005) |
| Married | 0.113*** | −0.020 | 0.083* |
| | (0.017) | (0.032) | (0.045) |
| Race | | | |
| African/Black | −0.353*** | −1.292*** | 0.936*** |
| | (0.037) | (0.144) | (0.167) |
| Coloured | −0.071** | −1.204*** | 0.920*** |
| | (0.044) | (0.141) | (0.164) |
| Indian/Asian | −0.079 | −0.510* | 0.365 |
| | (0.074) | (0.222) | (0.252) |
| White | - | | |
| Fetch water | −0.161*** | | |
| | (0.033) | | |
| PSU formal employment rate | 1.544*** | | |
| | (0.073) | | |
| Regional dummies | *Yes* | *Yes* | *Yes* |
| Constant | −0.182*** | | |
| | (0.077) | | |
| Number of observations | 37578 | 36640 | 36640 |

*Notes*: The selection equation uses Probit estimates (the positive outcome is formal employment) and the wage equation is based on double residual semi-parametric regression estimates. Significance is denoted by * for 10 per cent level, ** for 5 per cent level and *** for 1 per cent level; standard errors are clustered at the PSU level.

**Table 4.** Uganda estimates

| | Selection Equation | Wage Equation |
|---|---|---|
| Worker's Characteristic | Coefficient (Standard Error) | Coefficient (Standard Error) |
| Years of education | 0.092*** | 0.041* |
| | (0.014) | (0.024) |
| Female | 0.197*** | −0.185** |
| | (0.086) | (0.090) |
| Age | 0.167*** | 0.020*** |
| | (0.027) | (0.006) |
| Age squared | −0.002*** | −0.001** |
| | (0.000) | (0.000) |
| Married | 0.225** | 0.177* |
| | (0.096) | (0.103) |
| Number of elderly persons | 0.069** | |
| | (0.033) | |
| Community characteristics | | |
| Agricultural extension services | 0.198* | |
| | (0.123) | |
| Market selling agricultural inputs | 0.412* | |
| | (0.228) | |
| Regional and time fixed effects | *Yes* | *Yes* |
| Constant | −5.492*** | |
| | (0.563) | |
| Number of observations | 1669 | 746 |

*Notes*: As for Table 3.

In line with descriptive statistics and literature findings, younger/unexperienced workers are more likely to be employed informally while older and married workers are more likely to work formally. In South Africa, there are also race differences: a significant portion of the South African population is white, and whites participate in formal employment at higher rates than non-whites. This is partly due to the spatial separation between business centres and areas of residence of Africans (Banerjee et al., 2008) inherited from apartheid. Finally, several exclusion variables are used to provide exogenous variations of the propensity scores necessary to identify the $MTE_U$. They are data-specific and discussed in detail in the next subsection. The resulting distributions of the propensity scores and common supports of the formal and informal distributions are analysed in Online Appendix D.

### 5.2. Exclusion Variables

The independence condition (Online Appendix C) requires exclusion variables $Z \setminus X$[3] satisfying classical relevance and exogeneity validity assumptions: a) they should affect the probability of working formally versus informally and b) they should have no direct effect on workers' earnings.

The advantage of the large sample from South Africa is the division of the sample by a significant number of geographical areas, 3080 PSUs (primary sampling units). On average there are between 30 and 130 individuals living in a PSU. We use the PSU areas to create a within-PSU formal employment rate. The local formalisation rate relates directly to the level of barriers to entry into formal employment, $b_1$, determining function $U_D$ (5) of the selection equation unobservables. The first column of Table 3 shows that the within-PSU formalisation rate is a strong predictor of the formal/informal allocation. The corresponding estimate (1.544) is positive and significant at the 1 per cent significance level implying lower barriers to entry into formal employment at the localities with higher formalisation level (the corresponding elasticity at the average rate is 0.6). This validates the relevance of the within-PSU unemployment rate.

The exogeneity requirement would be violated if the PSU area operated as distinct markets with their own equilibrium wages. If it were the case, then within-PSU average wages would be defined within the areas and would be correlated with within-PSU formalisation rates. However, PSU-level regressions of average wages on within-PSU formalisation rates do not report correlations between the wages and formalisation rate at the PSU level (the estimates are reported in Table D1, Online Appendix D). It implies that the small size of the areas is insufficient for leaving them to operate as local markets.

The other exclusion variable, fetching water, is negatively correlated with the probability of holding a formal job. Its marginal effect is 6 per cent. Fetching water is likely to be negatively correlated with the infrastructure level of the region and therefore with the level of industrialisation and formalisation. The higher the level of formalisation, the higher the probability of working formally.

On the other hand, underdeveloped infrastructure can be a feature of localities with poorer populations and lower market wages. To check the irrelevance of the fetching water indicator as a covariate of income, PSU-level average income is regressed on within-PSU fetching water rates. The estimate (column 2 of Table D1) shows that the rate of fetching water is not correlated with PSU-level income when holding the formal employment rate as well as average human capital and demographic characteristics constant. This supports the exogenous nature of the indicator.

In the Ugandan case study, household structure variables have been explored as exclusion restrictions. The approach builds on Günther and Launov (2012), arguing that household-specific variables influence one's behaviour in the labour market but not one's earning potential. Indeed, household structure impacts one's budget and time constraints and consequently affects one's labour supply and occupational choice. Higher numbers of dependants can increase the pressure on both constraints. On the other hand, conditional on health status, old persons can contribute to the budget (for example by receiving pensions) or into household production and household care. In such a case, their presence in the households may alleviate the pressure and consequently reduce the job search costs, allow for higher mobility, higher effort in job search and engagement in activities requiring regular working hours. Household structure and intra-household transfers may therefore impact job search costs, reservation wages and time constraints while they have no direct impact on the earning potential of individuals.

Among household structure variables, the number of elderly persons is found to be positively correlated with one's likelihood to hold a formal job. The corresponding marginal effect is 1.5 per cent. Exogeneity of the variable is controversial if living in extended families is conditioned by the type of dwelling and therefore, household income and one's wage. To test robustness, the type of dwelling (having an independent house) is added as an explanatory variable in the probit regression. The estimates (reported in Table D2, Online Appendix D) show that the coefficient associated with the number of elderly is not altered, implying no endogeneity related to income. Another concern could be endogeneity related to the type of one's activity in the case where older and/or extended families are more likely to live in rural areas and thus one's occupation is more likely to be rural-specific. This would imply a correlation between the type of one's family and one's wage. To investigate this possibility, one's occupation is included in the list of control variables. Once again, the coefficient associated with elderly is robust, meaning no endogeneity related to one's occupation (Table D2, Online Appendix D).

In addition to household structure, the community part of the survey has been exploited. Availability of agricultural extension services and a market selling agricultural inputs increases the probability of formal employment. The corresponding marginal effects are 5 and 12 per cent. The likelihood ratio test rejects the null hypothesis of no relationship between one's community agricultural infrastructure and one's likelihood to work formally versus informally (the corresponding p-value is below 1%). Development of agricultural services and markets might signal communities with stronger administrative infrastructure involving agricultural, trade and service workers representing a large fraction of informal employment, and consequently higher local formalisation rates among them. However, if these services and markets are available in regions where other infrastructure is also more developed, their exogeneity is questionable: variation in formality could be confounded with other infrastructure, positively impacting local wages. To examine this threat, other services such as health centres, post offices and banks are added as controls. Neither effect on the likelihood of formal employment nor significant change in estimated effects of the exclusion variables is reported (Table D2, Online Appendix D).

The last piece of empirical evidence suggestive about exogeneity of the exclusion variables comes from the regressions of the residuals $\varepsilon$ of estimated models (A1, Online Appendix) on the exclusion variables: the coefficients associated with the instruments are not statistically significant and related $R2$ are close to zero (the last columns of Tables D1 and D2, Online Appendix). Independence condition also implies orthogonality between the unobservables of the selection equation, $U_D$, and the exclusion variables $Z \setminus X$[4]. The exclusion variables used represent mainly $b_1$, the levels of formalisation proxing barriers to entry into formal employment. Given model (5), the assumption translates into independence between $Z \setminus X$ and $(b_0, (U_1 - U_0))$.

Orthogonality between $Z \setminus X$ and $(U_1 - U_0)$ is consequent to independence of $Z \setminus X$ and $(U_1, U_0)$ discussed above. Independence between $Z \setminus X$ and $b_0$ relies on a plausible assumption that the entry barriers $b_0$ are specific to the inner functioning and development of informality and relate to unemployment rather than to the degree of formalisation relating to $b_1$.

### 5.3. Earnings from Formal versus Informal Employment

Along with the Egyptian study offered by Radchenko (2014), the results fit the three major scenarios discussed in Section 4. The central aggregate estimates of the treatment parameters for South Africa and Uganda are displayed in Table 5[5] and Figures 3 and 4.

South Africa estimates show negative sorting on the gain ($0<TT<ATE<TNT$) fitting the third pattern discussed in Section 4. All the effects are positive, implying higher returns from working formally for any worker. On average, the formal earning of a worker randomly drawn from the sample is more than twice as high as their informal counterfactual earnings. Roughly 50 per cent of the differential is related to different returns to observables (education, gender, race, and so forth). More than 100 per cent of the differential is related to unobservable characteristics.

The results suggest a negative sorting on the gain in terms of both observable and unobservable characteristics. Race and gender are among the factors of negative sorting related to observables (Table 3): Africans and women are less likely to work formally while having lower returns from informal employment. This is related to the historical legacy of apartheid, where Africans and in particular women were not allowed to work formally and had to hold subsistent jobs. Social stigma in

**Table 5.** *ATE, TT, TNT* and their decompositions

| Country | Treatment Effect | Total | $TT_X$ $ATE_X$ $TNT_X$ | $TT_U$ $ATE_U$ $TNT_U$ | $SB_{1\to 0}$ | $SB_{0\to 1}$ |
|---|---|---|---|---|---|---|
| South Africa | *TT* | 0.976*** (0.039) | 0.422 | 0.555*** | −0.168 | 1.146 |
| | *ATE* | 1.554*** (0.039) | 0.462 | 1.093*** | | |
| | *TNT* | 2.073*** (0.062) | 0.541 | 1.532*** | | |
| Uganda | *TT* | 1.340** (0.641) | −0.670 | 1.979*** | −1.269 | 0.553 |
| | *ATE* | 0.900** (0.425) | −0.793 | 1.657*** | | |
| | *TNT* | 0.473 (0.441) | −0.828 | 1.263*** | | |

*Notes*: Population effects are estimated using sample weights. Significance is denoted by * for 10 per cent level, ** for 5 per cent level and *** for 1 per cent level.

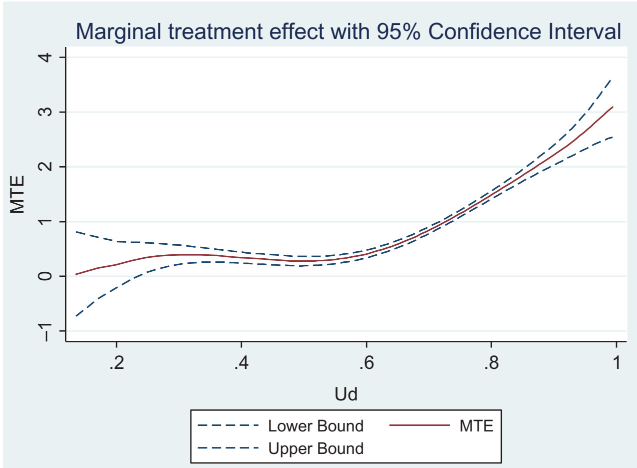


**Figure 3.** South Africa.

*Notes*: The propensity score is ordered by the horizontal axis. Low abscissa values correspond to the workers who are less likely to hold formal job than informal one based on their observable characteristics. As discussed in Section 3.1, $\mathrm{MTE_U}$ are evaluated at the indifference points where propensity scores equal $U_D$. Low abscissa values are associated therefore with high probability to be employed in the formal versus informal sector based on unobservable characteristics holding the observables fixed.

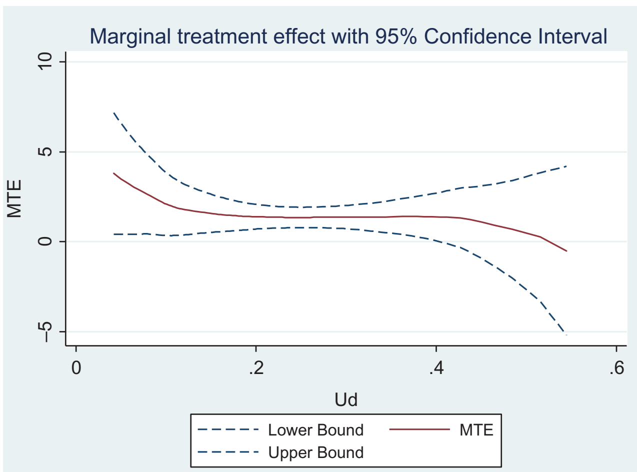


**Figure 4.** Uganda.

*Notes*: The propensity score is ordered by the horizontal axis. Low abscissa values correspond to the workers who are less likely to hold formal job than informal one based on their observable characteristics. As discussed in Section 3.1, $\mathrm{MTE_U}$ are evaluated at the indifference points where propensity scores equal $U_D$. Low abscissa values are associated therefore with high probability to be employed in the formal versus informal sector based on unobservable characteristics holding the observables fixed.

regards to women's work also induced women to engage in payed household-related work, which is more frequently informal and less remunerative (Valodia, 2001).

In terms of unobservable characteristics, the least educated portion of the African population (with some primary education at most) would not gain from working formally. This is shown by the flat shape of the $MTE_U$ curve at its left tail (Figure 3). Unskilled African workers do not sort themselves based on the anticipated individual gains. The unobserved factors pushing workers into informal employment might be rather specific to the local labour markets and limited occupational opportunities for unskilled workers as described by selection mechanism (8) in Section 3.2. Therefore, formal employment of unskilled workers is rationed and workers' allocation between the two employment groups is weakly inefficient in line with the second pattern presented in Section 4.2. Workers having at least some secondary education gain more if working formally. This is due to the higher returns to education in formal employment. Moreover, relative gains from formal employment are channelled through unobservable characteristics as implied by positive $MTE_U$. However, the upward shape of the $MTE_U$ curve implies negative sorting on the gain: the larger the individual earning gaps, the smaller the propensity for working formally based on unobservable characteristics.

The first of the scenarios fitting the negative sorting on the gain and offered in Section 4.3 is the prevalence of non-monetary comparative advantages of the informal jobs over their monetary advantages where monetary and non-monetary components are negatively correlated. The specific context of South Africa, with remarkably high unemployment, does not allow for the application of such a scenario. Strong unemployment along with an underdeveloped informal sector is a controversial issue in the literature on the South African labour market. On the one hand, the unemployed labour force might have relatively high reservation wages and prefer open unemployment over lower tier of informal employment (*Unemployment* $\succ$ *Lower tier*). In support of this, the OECD points to the widespread welfare system in South Africa. According to the QLFS data, only 15 per cent of the inactive population relies on old age pensions and 13 per cent receive child support grants; 77 per cent of both unemployed and inactive workers are supported instead by other members of their households. Household transfers might also increase the reservation wages of unemployed individuals. These data imply that at least a fraction of the unemployed labour force consists of workers unwilling to engage in the lower tier of informal employment.

On the other hand, the phenomenon is explained by increasing unskilled labour supply and an influx of female labour supply on the South African labour market over the transition period since the dismantling of apartheid (Banerjee et al., 2008). It is also explained by de-industrialisation of the South African economy: according to Rodrik (2008), the weakening of manufacturing and a shift to the tertiary sector yielded a drop in demand for low-skilled workers. A large influx of unskilled workers combined with shrinking demand for them yields an unlimited labour supply to the lower tier of the informal sector. For its poorest layer, unemployment is the worst option (*Lower tier* $\succ$ *Unemployment*). This body of literature implies a high fraction of the labour force willing to engage in the lower tier of informal employment and having low reservation wages (Kingdon & Knight, 2004).

Overall, the literature suggests a heterogeneous pool of job seekers. The results of the present analysis suggest that informal employment of the South African labour market hosts the poorest of them, those who have the lowest reservation wages. Indeed, the negative sorting on the gain implies that workers most likely to be pushed toward informal employment are those whose losses from working informally are particularly high. The high losses yield particularly low earnings from informal employment. Such a behavioural pattern is in line with the second scenario offered in Section 4.3 and signals the selection mechanism described by (9), suggesting barriers to entry not only into formal but also informal employment. The pattern implies a strong prevalence of the lower tier of informal employment over its upper tier: South African workers use informal jobs as a last resort and accept any wage level. It also implies inefficient allocation of workers in terms of their social welfare: as stated in Section 4.3, the higher losses from workers' observed allocation associated with their higher likelihood imply a negative mean gross surplus $S_A$.

The Ugandan estimates do not show different returns to the observable individual characteristics of formal and informal workers. The inefficiency of the differential estimates (not reported) may be due to an insufficient sample size. Yet, the wage equation following the classical Mincer's format shows positive return to education and experience proxied by age (Table 4). Positive and constant $MTE_U$ (Figure 4) implies higher earnings in the formal sector for any worker and no correlation between the gains and the selection process. The Ugandan estimates fit with the scenario discussed in Section 4.2. The results provide strong evidence of a segmented labour market in Uganda which pays higher wages in formal employment as framed by (8), Section 3.2. The aggregate effects are consistent with positive and homogenous $MTE_U$: they are positive and close in magnitude. More detailed interpretation of the aggregate effects would not be reliable as these effects are based on the weak or incomplete support

(Online Appendix D explores the propensity scores supports) and risk being subjected to bias.[6] Overall, the Ugandan results imply a weakly inefficient allocation of workers between formal and informal employment due to the strongly rationed formal employment; the embryonic state of the formal economy does not leave room for developing competitive mechanisms or shaping various tiers within informal employment.

## 6. Conclusions

This paper contributes to the literature on the heterogeneity of informal employment in developing economies. Drawing on the model with essential heterogeneity, it enriches the analysis offered by Radchenko (2014) by extending a list of scenarios describing behavioural patterns followed by informal workers. The paper shows that the model with essential heterogeneity nests not only classical patterns of a rationed formal sector versus an integrated labour market, but also different patterns of rationing. The analysis yields disparate impacts of rationing on the social welfare of informal employment.

The patterns are illustrated using non-parametric techniques and data from South Africa and Uganda, with comparisons to Egypt. These labour markets show heterogeneous observable and unobservable human capital and demographic characteristics of workers engaged in the formal and informal employment groups. The relative gains or losses from working formally rather than informally are heterogeneous in the cases where shares of formal and informal employment are balanced. They are homogenous in the case of Uganda, where the share of formal employment is very low.

Furthermore, the mechanisms inducing workers to work informally are heterogeneous. The South African results suggest inefficient rationing of formal employment and imply that workers use informal jobs as a subsistence strategy and a last resort to unemployment. African workers and women with low or no education are more likely to work informally though they would benefit more from working in the formal sector, which provides a higher return to education and exhibits less discrimination. The allocation of more educated workers and/or whites is strongly inefficient in terms of workers' social welfare: workers with stronger relative gains from working formally are more likely to work in the informal sector. Unlike the South African case, the rationing of the Ugandan formal employment is weakly inefficient exhibiting zero sorting on the gain.

Overall, the results show multiple heterogeneity of informal employment in African economies. Heterogeneity arises not only from the individual skills of formal and informal workers and individual gains from working formally versus informally, but also from the mechanisms inducing workers to work informally. Moreover, the results show different levels of inefficiency, implying that rationing of formal employment yields multifaceted impacts. Thus, the results raise a question on the suitability of such a rough classification as a dual market versus an integrated market, suggesting that there may be various schemes of rationing within the dual market and transitional forms between dual and integrated markets where the frontiers between two theories are hard to define.

The methodology is less appropriate for such economies as Uganda with low levels of development and small formal sectors (the relatively small size limits technical analysis). In spite of this, the results confirm the base intuition related to the low involvement in formal employment: the formal economy is so weakly developed that its constrained nature does not leave room for the development of competitive mechanisms and shaping of various tiers within informal employment. Paradoxically, the results imply that more advanced economies may represent strongly inefficient allocations of workers between the different types of employment, while less developed economies may retain weakly inefficient allocations.

The intra-country comparisons, while over a limited set of countries, illustrate different schemes of allocation of workers between formal and informal sectors, and more interestingly different inefficiency patterns in terms of the social welfare of workers. This result raises questions regarding the relationship between the efficiency of informal employment within a developing economy and the economy's level of development, unemployment rate, and degree of urbanisation. Future research should integrate unemployment and urbanisation into the model.

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## Notes

1. Agricultural employment is excluded.
2. Higher certificates of 1–3 additional after-secondary years of education or vocational school certificates.
3. Following conventional notations, $Z \backslash X$ denotes relative complement of the Z set, that is the variables included in the $Z$ set but excluded from the set of $X$ variables.
4. The assumption is conventionally admitted implicitly by empirical applications of the model with essential heterogeneity.
5. The results replicating the Egyptian case using the improved set of exclusion variables are reported in Online Appendix F.
6. Identification is particularly problematic for *ATE* and *TNT; TT* is identified provided the left tail of the support (See Online Appendix C, condition 5).

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